ASSIGNMENT 1

COURSE NAME	CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE	CSA5111
TOTAL MARKS	50 Marks
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Evaluate, Design and Defend a Secure Patient-Record Exchange Strategy

Course Outcome CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.

Total Marks: 50

Answer format: Each task question is reproduced first, followed immediately by its answer. The discussion follows the assignment scenario and rubric.

Design assumptions and notation

- The healthcare network contains one central hospital and 25 satellite clinics. The public network is assumed to be observable and vulnerable to interception.
- Security labels are ordered Public < Restricted < Confidential < Secret < Top Secret. Compartments are used in addition to the level, for example {CARDIOLOGY}, {ONCOLOGY}, or {BILLING}.
- For classical cipher demonstrations, letters are represented by A=0 to Z=25. The Hill-cipher extension uses A–Z followed by digits 0–9, giving modulus 36. Spaces and punctuation are removed before encryption and restored by a documented formatting rule.
- Classical ciphers in this assignment are educational demonstrations, not safe choices for real patient records. This limitation is considered explicitly in the final recommendation.

TASK 1 – Evaluate Threats, Attacks and Risks (4 Marks)

Question: Identify at least six security attacks relevant to the healthcare scenario. Classify each as passive or active; distinguish threat, vulnerability, attack and risk; rank the attacks using likelihood × impact; and defend which attack should be addressed first.

Answer

A threat is a potential cause of harm, such as a dishonest insider or an external attacker. A vulnerability is a weakness that makes the harm possible, such as a shared key or weak staff authentication. An attack is the actual attempt to exploit a vulnerability. Risk is the combination of the likelihood of the attack and the impact if it succeeds.

Attack	Type	Scenario example / weakness	Likelihood (1–5)	Impact (1–5)	Risk score
Eavesdropping	Passive	Attacker captures clinic-to-hospital ciphertext on the public network; weak classical ciphers may reveal diagnoses.	4	4	16 – High
Traffic analysis	Passive	Observer studies message timing, size and clinic destinations to infer emergencies or specialist visits.	4	3	12 – High
Prescription / lab-value modification	Active	Attacker changes dosage, allergy status or potassium value while a message is in transit; no strong integrity check.	3	5	15 – High
Masquerade as a doctor	Active	Attacker uses a stolen/shared password or impersonates a specialist and sends a false	3	5	15 – High

		order.			
Replay of an old order	Active	Previously valid prescription is resent because messages have no timestamp, sequence number or freshness check.	4	5	20 – Critical
Denial of service	Active	Records server or network is flooded or disabled, preventing access to allergy and treatment information.	3	5	15 – High
Insider ciphertext collection	Passive initially; may enable active attack	Clinic employee copies many encrypted messages under a reused key and exploits predictable headers.	4	5	20 – Critical

The scores are qualitative, not measured incident probabilities. Likelihood and impact are rated from 1 (very low) to 5 (very high); score = likelihood × impact. The insider collection and replay attacks receive the highest scores because repeated predictable messages can expose many records, while replay can directly cause unsafe treatment.

Priority: address replay and prescription integrity first, while also containing the insider key exposure. A replayed prescription can cause immediate patient harm even if confidentiality is preserved. The system should bind each order to a patient, prescriber, timestamp, unique sequence number and validity period, and staff must verify it before administration. Under the classical-only exercise, a checksum and strict process controls can detect some accidental changes, but an attacker who can recompute the checksum can defeat it.

TASK 2 – Map and Judge Security Goals (3 Marks)

Question: For record upload, prescription transmission and specialist consultation, decide which security goals matter most: confidentiality, integrity, availability, authentication and non-repudiation. Judge which goals classical ciphers can and cannot deliver. Explain why correct decryption does not prove the sender or message integrity.

Answer:

Task 2: Map and Judge Security Goals

The healthcare network must protect patient records during uploading, prescription transmission and specialist consultations. It requires five major security goals:

1. **Confidentiality:** Prevents unauthorized people from accessing patient information.
2. **Integrity:** Ensures medical records and prescriptions are not modified.
3. **Availability:** Ensures information is accessible whenever needed.
4. **Authentication:** Verifies the identity of doctors, nurses and other users.
5. **Non-repudiation:** Prevents users from denying actions they have performed.

Mapping Security Goals to System Functions:

System Function	Required Security Goals
Record Upload	Confidentiality, Integrity, Authentication, Availability
Prescription Transmission	Confidentiality, Integrity, Authentication, Non-repudiation
Specialist Consultation	Confidentiality, Integrity, Authentication, Availability

Limitations of Classical Ciphers:

- **Confidentiality:** Classical ciphers provide basic encryption but can be broken using frequency analysis and brute-force attacks.
- **Integrity:** They cannot detect whether encrypted information has been modified.
- **Availability:** Encryption does not ensure that medical records or services remain accessible.
- **Authentication:** They cannot independently verify the identity of the sender.
- **Non-repudiation:** They cannot prove who actually sent a message because multiple users may share the same key.

Example:

Suppose a doctor sends a prescription for **500 mg of medicine**. An attacker modifies the encrypted message so that it decrypts to **1000 mg**. Even if the receiver decrypts it correctly, this does not prove that the prescription came from the correct doctor or that its dosage was not modified.

Security Requirements Beyond Classical Encryption:

To overcome the limitations of classical encryption, the healthcare network requires additional security measures.

Security Requirement	Recommended Measure
Confidentiality	Encryption of patient records
Integrity	Checksum and message verification
Authentication	User identity and password verification

Availability	Backup systems and reliable network infrastructure
Non-repudiation	Digital signatures in a real-world deployment

Confidentiality, integrity, availability, authentication and non-repudiation are all important security goals in the healthcare network. Classical encryption techniques provide basic confidentiality but cannot independently guarantee the remaining security requirements.

A successfully decrypted message does not necessarily prove that it originated from the authorized doctor or that its contents were not modified.

Therefore, classical ciphers must be supported by appropriate access-control policies, identity verification, integrity-checking mechanisms and reliable system infrastructure to protect patient records and ensure safe healthcare communication.

TASK 3 – Apply and Critique the Bell–LaPadula Model (5 Marks)

Question: Define security levels and subject/object assignments. Apply the simple security property (no read up) and *-property (no write down) to at least five access requests, including two denials. Evaluate emergency access, lower-level billing notes, tranquility and trusted subjects. Discuss limitations and suitability.

Answer

Level	**Network role / typical objects**
Top Secret	Medical Director; highly sensitive executive or restricted clinical oversight records.
Secret	Specialist doctors; specialist diagnoses and treatment plans.
Confidential	Nurses and laboratory staff; nursing notes, observations and lab records.

Restricted	Billing and front-desk staff; invoices, appointment and administrative data.
Public	Patients; their own released summaries, appointment instructions and public information.

Bell–LaPadula (BLP) is a confidentiality model. The simple security property states “no read up”: a subject cannot read an object at a higher security level. The *-property states “no write down”: a subject cannot write information to a lower level, because this could leak higher-level information. In this example, a subject may generally read at or below their level and write at or above their level, subject to compartments and need-to-know.

Access request	**Decision**	**Reason**
Nurse (Confidential) reads Confidential allergy record	Allow	Same level; no read up violation, assuming matching care compartment.
Nurse (Confidential) reads Secret specialist allergy note	Deny	Would read up from Confidential to Secret.
Billing clerk (Restricted) reads Confidential diagnosis	Deny	Would read up; billing role does not need the diagnosis.
Specialist doctor (Secret) writes a Secret treatment note	Allow	Same level; no write down.
Specialist doctor (Secret) writes diagnosis text into Restricted invoice	Deny	Would write down and leak Secret information to Restricted.
Medical Director (Top Secret) reads Secret specialist record	Allow	Reading down is permitted by the simple security property, if compartment authorization exists.
Patient (Public) reads released	Allow	Object has been formally

Public discharge summary		declassified to Public.

Emergency access and trusted subjects

If a nurse urgently needs a Secret-level allergy record, ordinary BLP denies access. A controlled break-glass procedure may temporarily authorize emergency access: the nurse states a reason, identity is checked, access is time-limited, the event is logged, and a senior clinician reviews it. This is an explicit, audited exception rather than ordinary unrestricted access. The model itself does not decide whether an emergency is genuine.

A Secret doctor cannot directly write Secret diagnosis details into a Restricted billing file because that is a write-down. A trusted subject or declassification authority can create a carefully minimized billing statement, such as "specialist consultation completed" and the approved charge, without copying diagnosis details. The trusted subject must be narrowly authorized, accountable and audited.

Tranquility means security labels should not change unpredictably during operation. Strong tranquility keeps labels fixed; weak tranquility permits changes only under controlled rules. In healthcare, changing a record label during an emergency or releasing a summary requires an authorized procedure, not an informal user decision.

BLP is useful for confidentiality and least-privilege reasoning, but it does not guarantee integrity, accuracy, freshness or availability. It may also be rigid during emergencies, and covert channels can leak information through timing or resource use. Therefore BLP alone is not sufficient for healthcare. It must be combined with integrity controls, authentication, audit trails, availability planning and clinical emergency procedures.

TASK 4 – Design the Cryptographic Strategies (4 Marks)

Question: Select at least three meaningfully different classical schemes. For each, provide encryption/decryption algorithm or formula, key definition, a worked example on a 20+ character scenario message and a justification for including it.

Answer

Chosen schemes: Caesar substitution (simple monoalphabetic shift), Vigenère (polyalphabetic substitution), and columnar transposition. Hill cipher is also demonstrated in Tasks 5 and 9 for numeric identifiers and dosage strings.

A. Caesar cipher

Key: integer $k \in \{0,\ldots,25\}$. Encryption: $C_i = (P_i + k) \bmod 26$. Decryption: $P_i = (C_i - k) \bmod 26$. The same shift is applied to every letter.

Worked example (k = 3): "PATIENT RECORDS NEED SAFE TRANSFER" is normalized to PATIENTRECORDSNEEDSAFETRANSFER. Encrypting each letter three positions forward gives "SDWLHQWUHFRUGVQHHGVDIHWUDQVIHU" (spaces removed). Decryption shifts each ciphertext letter three positions backward and restores the original text.

Reason for selection: Caesar is easy to implement and explains substitution and brute-force search. Its weakness is also useful for comparison: only 25 non-trivial keys exist.

B. Vigenère cipher

Key: a repeating keyword, for example MED. Map A=0,…,Z=25. Encryption: $C_i = (P_i + K_i) \bmod 26$; decryption: $P_i = (C_i - K_i) \bmod 26$. The key is repeated to match the plaintext length.

Worked example: plaintext PATIENTRECORDSNEEDSAFETRANSFER; key MED repeated to the same length. For the first three letters, P=A? Specifically P(15), A(0), T(19) and key M(12), E(4), D(3) produce ciphertext B(1), E(4), W(22), so PAT begins BEW. Apply the same addition to each remaining character. Decryption subtracts the repeated key values modulo 26.

Reason for selection: Vigenère is polyalphabetic and changes the substitution according to key position. It is stronger than a single fixed substitution, but a short repeating key creates periodic patterns that Kasiski examination and index of coincidence can expose.

C. Columnar transposition

Key: a keyword that determines the order in which columns are read. Write the plaintext row-wise under the keyword, pad the last row if needed, and read columns in ascending key-letter order. Decryption reconstructs columns in that order and reads rows.

Worked example: use keyword HOSPITAL and plaintext PATIENTRECORDSNEEDSAFETRANSFER (30 letters). Write the letters row-wise into eight

columns; the final row is padded with X characters. Rank the keyword letters alphabetically (A, H, I, L, O, P, S, T), read columns in that order, and concatenate them. Decryption fills the columns in the same order and reads row-wise, removing only the known padding.

Reason for selection: transposition rearranges positions without changing letter counts. It provides a useful contrast to substitution and demonstrates why frequency histograms alone do not identify a transposition key.

TASK 5 – Mathematical Analysis of the Schemes (5 Marks)

Question: Derive key-space sizes; show Hill inverse conditions and compute an inverse; express encryption as modular functions; and judge whether a large key space guarantees strength.

Answer

Scheme	Key-space size / formula	Assessment
Caesar	26 shifts including identity; 25 useful non-zero shifts.	Tiny; all keys can be tried instantly.
Monoalphabetic substitution	$26! \approx 4.03 \times 10^{26}$ permutations.	Large nominal space, but language frequencies and word patterns make it weak.
Vigenère, key length m	26^m possible strings (or $26^m - 1$ if excluding all-A key).	Strength depends on key length and non-repetition; repeating short keys are vulnerable.
Columnar transposition, n columns	n! column orders (some practical keys may be equivalent under padding/conventions).	Position rearrangement only; language structure remains exploitable.
Hill, n×n mod 26	Number of invertible matrices is $\|GL(n,Z_{26})\| = \|GL(n,F_2)\| \times$	Linear algebra structure permits known-plaintext recovery when

	$\lvert GL(n,F_{13})\rvert = \prod_{i=0}^{n-1}(2^n-2^i) \times \prod_{i=0}^{n-1}(13^n-13^i)$.	enough independent blocks are available.

For the Hill cipher over modulus 26, a key matrix K is invertible only when gcd(det(K),26)=1. Since 26=2×13, the determinant must be divisible by neither 2 nor 13. Encryption is C = KP mod 26 and decryption is $P = K^{-1}C \bmod 26$.

Worked inverse (3×3): choose K = [[1,2,3],[0,1,4],[5,6,0]]. Its determinant is 1. Therefore gcd(1,26)=1 and the matrix is invertible. Its adjugate/inverse modulo 26 is:

```
K^(-1) mod 26 = [[ 2, 18,  5],
                 [20, 11, 22],
                 [21,  4,  1]]
```

The modular inverse above is obtained by reducing the integer inverse modulo 26. Multiplying K by this matrix gives the identity matrix modulo 26.

For the requested mod-36 alphabet (A–Z, 0–9), the same condition becomes gcd(det(K),36)=1. Using K = [[1,2,3],[0,1,4],[5,6,0]], det(K)=1, so it is also invertible modulo 36. Its inverse modulo 36 is:

```
K^(-1) mod 36 = [[12, 18,  5],\n                 [20, 21, 32],\n                 [31,  4,  1]]
```

Large key space does not automatically mean strong security. Monoalphabetic substitution has 26! possible keys, but every plaintext letter always maps to the same ciphertext letter. English frequency peaks, common digrams and repeated word patterns remain visible. An attacker can use frequency analysis rather than search all 26! keys. Effective security depends on the structure of the cipher, key use, message volume and attacker knowledge—not only the number of possible keys.

TASK 6 – Evaluate Cryptanalytic Resistance (4 Marks)

Question: Attack each chosen scheme using a suitable technique (brute force, frequency analysis, Kasiski/IC, known-plaintext Hill attack, digram analysis). Explain why attacks succeed or fail and rank schemes by resistance.

Answer

Scheme	Suitable attack	Why it works / expected result
Caesar	Brute force (25 shifts) and frequency analysis	Try every shift and compare readable output; the search is negligible.
Monoalphabetic	Single-letter frequency, digram and word-pattern analysis	Substitution preserves frequency counts and repeated patterns; common E, T, A and TH/HE patterns guide recovery.
Vigenère	Kasiski examination and index of coincidence	Repeated key causes periodic ciphertext patterns. Repeated sequences suggest key-length factors; splitting by key position produces Caesar-like streams.
Columnar transposition	Anagram/column-order search and digram analysis	Letter counts are unchanged. Candidate column orders can be scored for likely language digrams and words.
Hill	Known-plaintext linear algebra	With enough independent plaintext/ciphertext blocks, solve C=KP mod m for K; linearity exposes the matrix.

For a Hill known-plaintext attack, collect three linearly independent plaintext column vectors P_1,P_2,P_3 and their ciphertext vectors C_1,C_2,C_3. Form $P=[P_1\ P_2\ P_3]$ and $C=[C_1\ C_2\ C_3]$. If P is invertible modulo the chosen modulus, $K = CP^{-1} \bmod m$. If P is not invertible, collect more blocks and select an invertible block matrix. This is a key-recovery method, not merely a guess.

Vigenère is generally more resistant than Caesar and monoalphabetic substitution when the key is long and not repeated; however, a short repeated key is vulnerable. Hill hides simple single-letter frequency

better because each output symbol depends on a block, but its linearity makes known-plaintext recovery practical. Columnar transposition preserves frequencies exactly and is vulnerable to structural analysis.

Overall educational ranking for the selected configurations: Caesar (weakest) < monoalphabetic substitution < simple columnar transposition < short-key repeating Vigenère < Hill against ciphertext-only frequency analysis. This is not a universal ranking: Hill can fall rapidly to known-plaintext attacks, and a long non-repeating Vigenère key can outperform these examples. No classical scheme here is suitable for real clinical confidentiality.

TASK 7 – Assess Transposition and Product Ciphers (3 Marks)

Question: Implement a columnar/rail-fence transposition and explain diffusion. Evaluate substitution followed by transposition against either alone, with frequency evidence and an attack attempt. Judge practical complexity for rural clinics.

Answer

1. Columnar Transposition:

- In columnar transposition, plaintext is written row-wise into columns and read column-wise according to a keyword.
- It rearranges the positions of characters without changing the actual letters.
- This spreads nearby plaintext characters across different ciphertext positions.
- However, the frequency of each letter remains unchanged, making it vulnerable to frequency analysis.

2. Product Cipher:

- A product cipher combines two techniques: substitution followed by transposition.
- Substitution replaces plaintext letters with different letters, while transposition rearranges their positions.
- This combination provides better security than using either technique alone.
- The frequency pattern changes after substitution, and transposition further hides the original letter arrangement.
- However, it is still vulnerable to frequency analysis, repeated-key attacks and other statistical methods.

3. Attack Attempt:

- In transposition alone, the frequency of each letter remains exactly the same as in the plaintext. An attacker can use this to identify the technique.
- In a product cipher, an attacker can examine letter frequencies to estimate the substitution pattern.
- The attacker can then use common word patterns and digram frequencies to guess the transposition order.

4. Practical Complexity for Rural Clinics:

- Manual encryption using multiple steps increases the chances of key-entry errors.
- It may also delay access to important patient information during emergencies.
- A printed key card and automated local software can make the process easier.
- Nurses and healthcare workers should not be required to perform complex encryption manually.

Product ciphers provide better protection than individual classical encryption techniques because they combine substitution and transposition. However, they are not strong enough for modern healthcare systems and should only be used for educational demonstrations.

TASK 8 – Evaluate Steganography

Question: Design a steganographic method for a short report inside a medical image. Compare it with encryption on capacity, imperceptibility, robustness and detectability. Assess diagnostic risk, appropriateness and encrypt-then-embed.

Answer

LSB image embedding stores each message bit in the least significant bit of a pixel channel. For a grayscale 256×256 image, there are 65,536 pixels. Using one LSB per pixel gives a theoretical capacity of 65,536 bits = 8,192 bytes before metadata and error-correction overhead. A short report can be embedded after adding a length header.

```
Embedding outline:\n1. Convert report to bytes and then bits.\n2. Store message length in a fixed header.\n3. For each bit, replace pixel LSB: pixel = (pixel & 254) | bit.\n4. Extract by reading LSBs in the same pixel order; rebuild bytes using the stored length.
```

Criterion	Encryption	LSB steganography
Purpose	Makes content unreadable without a key.	Conceals the presence of a message.
Capacity	Depends on algorithm and	Limited by available pixels and

	message size; no image capacity limit.	number of LSBs used.
Imperceptibility	Not applicable to the image appearance if file is separate.	Small LSB changes may be visually imperceptible.
Robustness	Ciphertext can be copied without changing meaning; integrity still needs a separate control.	Fragile: JPEG compression, resizing, filtering or format conversion may destroy hidden bits.
Detectability	Ciphertext is visibly encrypted data.	Statistical tests, including chi-square analysis, may detect altered LSB distributions.

Medical imaging requires special caution. Even if a one-bit change is visually invisible, pixel alteration can affect quantitative measurements or downstream diagnostic processing. Original diagnostic pixels should therefore remain unchanged and authoritative. Store hidden data only in a separately designated derivative/copy, preserve the original, and validate the image with clinical staff.

Steganography is appropriate only for low-risk metadata or a demonstration where policy permits it. It is not a substitute for encryption: if the hidden message is discovered, plaintext is exposed. The safer conceptual sequence is encrypt-then-embed: encrypt the report first, then embed ciphertext.

However, JPEG compression may destroy LSB data, so a robust transport channel and explicit extraction verification are required. For real patient data, use approved secure image metadata/transport standards rather than covert embedding.

TASK 9 – Implement and Test

Question: Implement the chosen schemes, cryptanalysis tools and feasible steganography. Provide correctness tests including digits, punctuation and Hill block padding. Test identical plaintexts of 100, 1,000, 10,000 and 100,000 characters and document a reproducible procedure.

Answer

The following Python reference implementation is designed for reproducible classroom testing. It includes Caesar, Vigenère, columnar transposition, Hill mod 36, and LSB embed/extract. It normalizes letters and digits for Hill; punctuation is preserved by the outer test harness or stored separately. For Hill, X padding is added to complete the final three-character block and removed after decryption.

```
import time, random, string
import numpy as np
from PIL import Image

ALPH = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
A36 = "ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789"
K36 = np.array([[1,2,3],[0,1,4],[5,6,0]], dtype=int)
K36_INV = np.array([[12,18,5],[20,21,32],[31,4,1]], dtype=int)

def clean_letters(s):
   return ''.join(c for c in s.upper() if c in ALPH)

def caesar_enc(s, k=3):
   return ''.join(ALPH[(ALPH.index(c)+k)%26] for c in clean_letters(s))
def caesar_dec(s, k=3):
   return ''.join(ALPH[(ALPH.index(c)-k)%26] for c in s)

def vig_enc(s, key="MED"):
   p=clean_letters(s); key=clean_letters(key)
   return ''.join(ALPH[(ALPH.index(c)+ALPH.index(key[i%len(key)]))%26]
                  for i,c in enumerate(p))
def vig_dec(s, key="MED"):
   key=clean_letters(key)
   return ''.join(ALPH[(ALPH.index(c)-ALPH.index(key[i%len(key)]))%26]
                  for i,c in enumerate(s))

def col_enc(s, key="HOSPITAL"):
   p=clean_letters(s); n=len(key)
   rows=[p[i:i+n].ljust(n,'X') for i in range(0,len(p),n)]
   order=sorted(range(n), key=lambda i:(key[i],i))
   return ''.join(row[j] for j in order for row in rows)
def col_dec(c, key="HOSPITAL", original_len=None):
   n=len(key); r=(len(c)+n-1)//n
   order=sorted(range(n), key=lambda i:(key[i],i))
   cols=['']*n; pos=0
   for j in order:
      cols[j]=c[pos:pos+r]; pos+=r
   out=''.join(cols[j][i] for i in range(r) for j in range(n))
   return out[:original_len] if original_len is not None else out.rstrip('X')
```

```
def hill36(s, matrix):
    x=''.join(c for c in s.upper() if c in A36)
    pad=(-len(x))%3; x += 'X'*pad
    out=''
    for i in range(0,len(x),3):
        v=np.array([A36.index(c) for c in x[i:i+3]])
        z=matrix.dot(v)%36
        out += ''.join(A36[int(a)] for a in z)
    return out, pad
def hill36_dec(c, pad=0):
    out=''
    for i in range(0,len(c),3):
        v=np.array([A36.index(ch) for ch in c[i:i+3]])
        z=K36_INV.dot(v)%36
        out += ''.join(A36[int(a)] for a in z)
    return out[:-pad] if pad else out

def lsb_embed(gray_array, message):
    bits=''.join(f'{b:08b}' for b in message.encode())
    header=f'{len(bits):032b}'
    stream=header+bits
    flat=gray_array.copy().astype(np.uint8).ravel()
    if len(stream)>len(flat): raise ValueError("Image capacity exceeded")
    for i,bit in enumerate(stream):
        flat[i]=(int(flat[i]) & 254) | int(bit)
    return flat.reshape(gray_array.shape)
def lsb_extract(stego):
    flat=stego.ravel()
    n=int(''.join(str(int(x)&1) for x in flat[:32]),2)
    bits=''.join(str(int(x)&1) for x in flat[32:32+n])
    data=bytes(int(bits[i:i+8],2) for i in range(0,n,8))
    return data.decode()

# Correctness tests
sample="PATIENT RECORDS NEED SAFE TRANSFER"
assert caesar_dec(caesar_enc(sample)) == clean_letters(sample)
assert vig_dec(vig_enc(sample)) == clean_letters(sample)
assert col_dec(col_enc(sample), original_len=len(clean_letters(sample))) == clean_letters(sample)
msg="PT20481AMOX500MG"
ct,pad=hill36(msg,K36)
assert hill36_dec(ct,pad) == msg

# Size/timing experiment; identical alphabetic plaintext per size
random.seed(7)
for n in (100,1000,10000,100000):
    text=''.join(random.choice(ALPH) for _ in range(n))
```

```
for name,enc,dec in [
    ("Caesar",caesar_enc,caesar_dec),
    ("Vigenere",vig_enc,vig_dec),
    ("Columnar",col_enc,lambda x: col_dec(x,original_len=n))]:
    t0=time.perf_counter(); c=enc(text); t1=time.perf_counter()
    p=dec(c); t2=time.perf_counter()
    assert p==text
    print(n,name,"enc",t1-t0,"dec",t2-t1)
```

Implementation notes:

1. install Python 3, NumPy and Pillow (`pip install numpy pillow`).
2. Save the code as `cns_classical.py` and run `python cns_classical.py`.
3. For image testing, load a grayscale PNG using Pillow, embed a short test string, save as PNG, reload and extract.
4. Do not use JPEG for the primary correctness test because JPEG is lossy.

Edge cases: the test includes digits in PT20481AMOX500MG and pads incomplete Hill blocks. The Hill input alphabet excludes hyphens and spaces; preserve those separators as external formatting metadata and reinsert them after decryption. For arbitrary Unicode or punctuation, encode and protect the original bytes/format separately rather than silently discarding characters.

The code is a reference demonstration, not a production cryptographic library. The Hill matrix inverse is fixed for modulus 36; changing the matrix requires recomputing and validating its modular inverse.

TASK 10 – Record and Interpret Experimental Evidence (3 Marks)

Question: Present tables and graphs for plaintext/ciphertext letter-frequency histograms, index of coincidence, encryption/decryption time against message size, and attack success rate against ciphertext length. Evaluate the data.

Answer

The experiment should use the same generated plaintext for each cipher at each length (100, 1,000, 10,000 and 100,000 characters). Use a fixed random seed so another student can reproduce the sequence. Record the actual run-time values from the computer used; timing depends on hardware, Python version and implementation.

Measure	How to calculate / record	Expected interpretation
Letter-frequency histogram	Count A–Z in plaintext and each ciphertext; plot counts or normalized percentages.	Caesar preserves the frequency profile with labels shifted; monoalphabetic also preserves the profile up to relabeling; transposition preserves exact counts; Vigenère tends to flatten the profile as key positions mix shifts.
Index of coincidence (IC)	$IC = \Sigma f_i(f_i-1) / [N(N-1)]$, where f_i is count of letter i and N is total letters.	English plaintext is commonly around 0.066; uniform random text is around 0.038. Caesar and monoalphabetic substitution preserve IC; transposition preserves it exactly; Vigenère often lowers it toward random depending on key length.
Encryption/decryption time	Use time.perf_counter() before and after each operation; report seconds for each input size.	Runtime should increase approximately linearly for simple character-by-character methods; columnar transposition includes sorting and matrix handling overhead.
Attack success rate	For each length, test a defined number of ciphertexts and record recovered keys/plaintexts divided by attempts × 100%.	Longer ciphertext generally provides more statistics and improves frequency-based attacks; a short repeated Vigenère key becomes easier to estimate as length grows.

For a fair attack-success experiment, define the attack algorithm and success criterion before testing. Example: run Caesar brute force on 100 ciphertexts per length and count a success only when the recovered plaintext exactly matches the original. For Vigenère, test candidate key lengths using

Kasiski/IC and record exact-key recovery. Do not label an attack “successful” merely because its output looks readable.

The supplied implementation prints timing measurements but does not claim measured values in this document. Run it on the same machine, paste the observed output into a results table, and create four graphs:

(1) plaintext/ciphertext frequency curves,

(2) IC by scheme,

(3) encryption/decryption time versus message size, and

(4) attack success percentage versus ciphertext length. This avoids presenting invented experimental measurements.

Expected interpretation: transposition does not change the histogram or IC because it only reorders characters. Monoalphabetic substitution changes which letter has each frequency but preserves the frequency distribution. Vigenère can reduce visible single-letter frequency peaks, but repeated-key periodicity remains. Larger ciphertexts often help an attacker estimate statistics more accurately. Thus, empirical results should be interpreted together with the mathematical weaknesses discussed in Tasks 5 and 6.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

Question: Recommend one scheme or layered combination with Bell–LaPadula. Justify against alternatives using mathematical and experimental evidence. State residual risks and limitations, judge whether classical ciphers are acceptable for patient data, and address key distribution across 25 clinics and usability.

Answer

For the assignment’s classical-technique comparison, the most defensible educational design is a layered demonstration using Vigenère with a sufficiently long key, followed by columnar transposition, together with Bell–LaPadula access control. Hill may be used to demonstrate block processing for numeric identifiers, but it should not be treated as a secure production solution.

- Confidentiality: Vigenère changes the substitution by key position and transposition rearranges the result. This is stronger than Caesar or monoalphabetic substitution alone, but repeated-key Vigenère and transposition remain vulnerable to known classical attacks.
- Access control: BLP labels restrict who may read and write records. Compartment labels such as {CARDIOLOGY} or {ONCOLOGY} ensure that clearance alone does not grant access to every clinical record.
- Integrity and freshness: attach a record identifier, timestamp, sequence number and independently checked checksum; verify prescription details with the prescriber. A classical checksum is not a secure message authentication code and can be forged by an informed attacker.
- Availability: maintain offline emergency access to essential allergy information, backups, redundant links and a documented downtime procedure.

Key distribution across 25 clinics is a major weakness. A single network-wide key would allow one compromised clinic to expose every clinic's messages. The classroom design should use a different key per clinic and rotate keys periodically, with a secure in-person or otherwise controlled handover process, a written inventory, named custodians and immediate revocation after staff departure or suspected compromise. Non-technical staff should use a simple approved interface rather than manually selecting ciphers or typing keys.

Residual risks include insider copying, key theft, frequency and known-plaintext attacks, replay, forged checksums, traffic analysis, denial of service, human error and emergency delays. Classical ciphers cannot provide modern authenticated encryption, robust digital signatures, strong non-repudiation or dependable integrity. Therefore classical ciphers are not acceptable as the actual protection for real patient data. A real deployment needs modern vetted cryptography, authenticated transport, digital identity and signatures, secure key management, audit logging, access reviews, backups and legal/clinical governance. This recommendation satisfies the learning objective while being honest about the boundary between a classroom design and a safe healthcare system.

TASK 12 – Day-4 Design Challenge: Re-evaluate and Adapt the Design (8 Marks)

Question: A clinic insider has copied hundreds of ciphertexts under the same key, many with predictable headers, and the network wants reports hidden inside

medical scan images. Re-evaluate the threat model, test the original scheme, decide whether to retain/modify/redesign, and defend the decision using key-space mathematics, evidence, usability and limitations.

Answer

A. Impact analysis (2 Marks)

The attacker is no longer only an unknown external observer. The attacker is an authorized insider at a satellite clinic who can collect many ciphertexts and may know the message format. This changes the assumptions: key secrecy at every clinic has failed; key reuse has exposed many samples under the same key; and predictable headers such as PATIENT ID, DIAGNOSIS and PRESCRIPTION provide known-plaintext or chosen-format clues.

A high volume of ciphertexts increases the value of frequency analysis and repeated-pattern analysis. For a repeating Vigenère key, repeated sequences and index-of-coincidence measurements become more reliable as the number and length of samples increase. Risk ranking therefore moves insider key compromise, known-plaintext recovery, forged/replayed prescriptions and unauthorized disclosure to the top.

B. Evaluation of the original design (2 Marks)

The original classroom combination—repeating-key Vigenère followed by columnar transposition—does not withstand the new attacker well. Vigenère key repetition creates periodic structure. If the same key is reused across hundreds of messages, the attacker can pool ciphertext statistics and exploit repeated headers. Columnar transposition changes positions but preserves symbol counts; it does not remove the statistical information in the collection.

Demonstration: if the known plaintext begins with PATIENTID and the corresponding ciphertext prefix is available before transposition, each aligned character gives a Vigenère key character by $K_i=(C_i-P_i) \bmod 26$. If transposition is applied after Vigenère, direct positions are disrupted, but repeated known headers across many messages still provide constraints on likely column order and substitution/key period. Once enough aligned plaintext/ciphertext blocks are recovered, key candidates can be tested against the remaining messages. This is a structural weakness, not a shortage of key-space size.

BLP still helps prevent an ordinary Restricted billing user from reading Secret records, and it limits permitted information flow. It does not stop an insider who is legitimately authorized to access and

copy ciphertexts, nor does it protect a key after that key is stolen. BLP governs subject/object access; it is not encryption, key management or integrity protection.

C. Adaptation or redesign (2 Marks)

Decision: modify the classroom design and redesign the operational controls. Do not retain a single shared key across all clinics.

- Use per-clinic keys so compromise at one clinic does not automatically expose every other clinic. Where possible, use per-session or per-message keys derived through a controlled key-management process; classical methods do not themselves provide a safe modern key exchange.
- Rotate and revoke keys after suspected exposure. Maintain a key register, restrict key custodians, and avoid distributing one universal key to all 25 sites.
- Remove predictable plaintext headers from the encrypted body where the receiving system can obtain routing metadata separately. Add randomised padding or standardised length buckets to reduce direct format and length leakage; this reduces some pattern leakage but does not repair weak cipher mathematics.
- Avoid reuse of short Vigenère keys. A truly random, message-length key used once would resist classical statistical attacks in theory, but distributing and securely storing such one-time keys for every message across 25 clinics is operationally impractical.
- For images, encrypt the short report first and only then embed the ciphertext in a designated non-diagnostic image copy. Keep the original scan pixel data unchanged and maintain a separate record of the embedded payload. Verify extraction after transfer; do not assume LSB survives JPEG compression.

Per-clinic keys increase the number of managed keys from one network key to at least 25 clinic keys. This reduces the blast radius but adds provisioning, rotation and revocation work. Per-session keys increase key-management effort further. Random padding increases message size and processing slightly, while removing fixed headers reduces predictable known-plaintext content. None of these changes makes a classical cipher equivalent to modern authenticated encryption.

D. Evidence-based defence (2 Marks)

The Hill example shows that an invertible matrix can be recovered using enough independent known plaintext/ciphertext blocks because $C=KP$ is linear. The Vigenère example shows key-space growth of 26^m, but a repeated key leaks periodicity, so nominal key-space size alone is not enough. Transposition preserves exact letter counts, so it does not defeat frequency analysis. These mathematical facts explain why the copied ciphertext collection materially weakens the original design.

The implementation and experiment plan in Tasks 9–10 should be used to record frequency histograms, IC values, timing and attack success at increasing ciphertext lengths. The expected evidence is that more samples improve statistical estimates and that transposition leaves frequency counts unchanged. Actual numerical measurements must be taken from the student's run rather than invented.

Final defence: use BLP for confidentiality-oriented role and clearance restrictions, per-clinic separation and controlled key rotation to reduce insider blast radius, and encrypt-then-embed only for an approved derivative image. Make the interface simple enough that clinic staff do not handle cipher calculations. Nevertheless, a real healthcare network must replace classical ciphers with modern authenticated encryption and digital signatures, secure key distribution, replay protection, immutable audit logs and clinical break-glass governance. Classical techniques cannot guarantee integrity, authentication, non-repudiation, resistance to modern computation or safe handling of compromised insiders.

Conclusion

The assignment demonstrates that classical encryption and Bell–LaPadula can explain important security principles, but they leave serious gaps. Substitution, transposition, Vigenère and Hill have exploitable statistical or algebraic structures. BLP controls information flow by security level but does not ensure that a record is correct or that a sender is genuine. For the healthcare scenario, patient safety requires confidentiality, integrity, authentication, freshness, availability and accountability together.